



Sizing Document

Manufacturing ERP System

Date: January 9, 2026

Client: Manufacturer

Estimated Duration: ~3 Weeks

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1) Executive Summary

This document provides resource sizing recommendations for deploying the Manufacturing ERP System on Azure. The sizing is focused on cost efficiency while maintaining scalability for future production workloads. It covers the required compute, network, storage, and global service resources to meet the needs of a demo environment.

2) Compute Resources

SQL Server Virtual Machine

- **VM Size:** Standard_B2als_v2
- **vCPUs:** 2, **Memory:** 4 GB
- **OS Disk:** 128 GB Premium SSD
- **SQL Version:** SQL Server 2022 Developer (for dev)
- **Cost (Demo):** ~\$45/month
- **Performance:** Burstable CPU, 3,200 IOPS, 80 MBps throughput.
- **Upgrade Path:** Options for Small, Medium, and Large Production workloads.

IIS Web Server Virtual Machine

- **VM Size:** Standard_B2als_v2
- **vCPUs:** 2, **Memory:** 4 GB
- **OS Disk:** 128 GB Standard SSD
- **Web App:** IIS 10, Python 3.11 + Flask
- **Cost (Demo):** ~\$45/month
- **Performance:** Handles 100-500 requests/sec, burstable CPU.



3) Network Resources

Azure Firewall

- **Primary Region:** Standard SKU, supports 30 Gbps throughput and 4,000+ SNAT ports
- **DR Region:** Same configuration as primary
- **Total Monthly Cost:** ~\$360/month
- **Traffic Analysis:** 10-50 Mbps typical, 2 DNAT rules, and WAF support.

VPN Gateway

- **SKU:** VpnGw1 (for P2S VPN, supports 5 admins)
- **Cost (Demo):** ~\$140/month
- **Bandwidth:** 650 Mbps, sufficient for admin traffic.

4) Storage Resources

Primary Storage Account (Italy North)

- **Tier:** Standard LRS (Hot tier)
- **Replication:** LRS
- **Cost (Demo):** ~\$2/month for 10-50 GB of Azure Files and Blob storage

DR Storage Account (North Europe)

- **Tier:** Standard LRS (Hot tier)
- **Cost:** ~\$17-20/month for replication traffic and blob versioning
- **Capacity Planning:** Separate LRS for DR, syncing data between Italy North and North Europe.



5) Global Services

Azure Front Door

- **Tier:** Standard, with WAF included
- **Cost (Demo):** ~\$40/month
- **Traffic:** ~10,000-50,000 requests/month, 5-20 GB of data transfer.

Recovery Services (ASR)

- **Cost (Demo):** ~\$56/month for protecting SQL + Web VMs in DR region.

6) Cost Summary

Monthly Cost Breakdown

- **SQL Server VM (B2als_v2):** ~\$45
- **Web Server VM (B2als_v2):** ~\$35
- **Azure Firewall (Primary + DR):** ~\$360
- **VPN Gateway (VpnGw1):** ~\$140
- **Azure Front Door (Standard):** ~\$40
- **Storage Account (Standard LRS):** ~\$2
- **ASR Protection:** ~\$56
- **Total Monthly Cost:** ~\$691
- **Total Daily Cost:** ~\$23

Production Estimate

- **Production Total:** ~\$1,184/month



Appendix A: VM Size Reference

Series	Use Case	Characteristics
B-series	Burstable	Low baseline, burst to 100%
D-series	General	Balanced compute
E-series	Memory	High memory ratio
F-series	Compute	High CPU ratio

Appendix B: Azure Firewall SKU Comparison

Feature	Basic	Standard	Premium
FQDN filtering	☑	☑	☑
Network rules	☑	☑	☑
Threat intelligence	✗	☑	☑
TLS inspection	✗	✗	☑
IDPS	✗	✗	☑
URL filtering	✗	☑	☑

Appendix C: Storage Performance Tiers

Tier	Max IOPS	Max Throughput	Latency
Standard HDD	500	60 MBps	Variable
Standard SSD	6,000	750 MBps	<10ms
Premium SSD	20,000	900 MBps	<1ms
Ultra Disk	160,000	4,000 MBps	<1ms